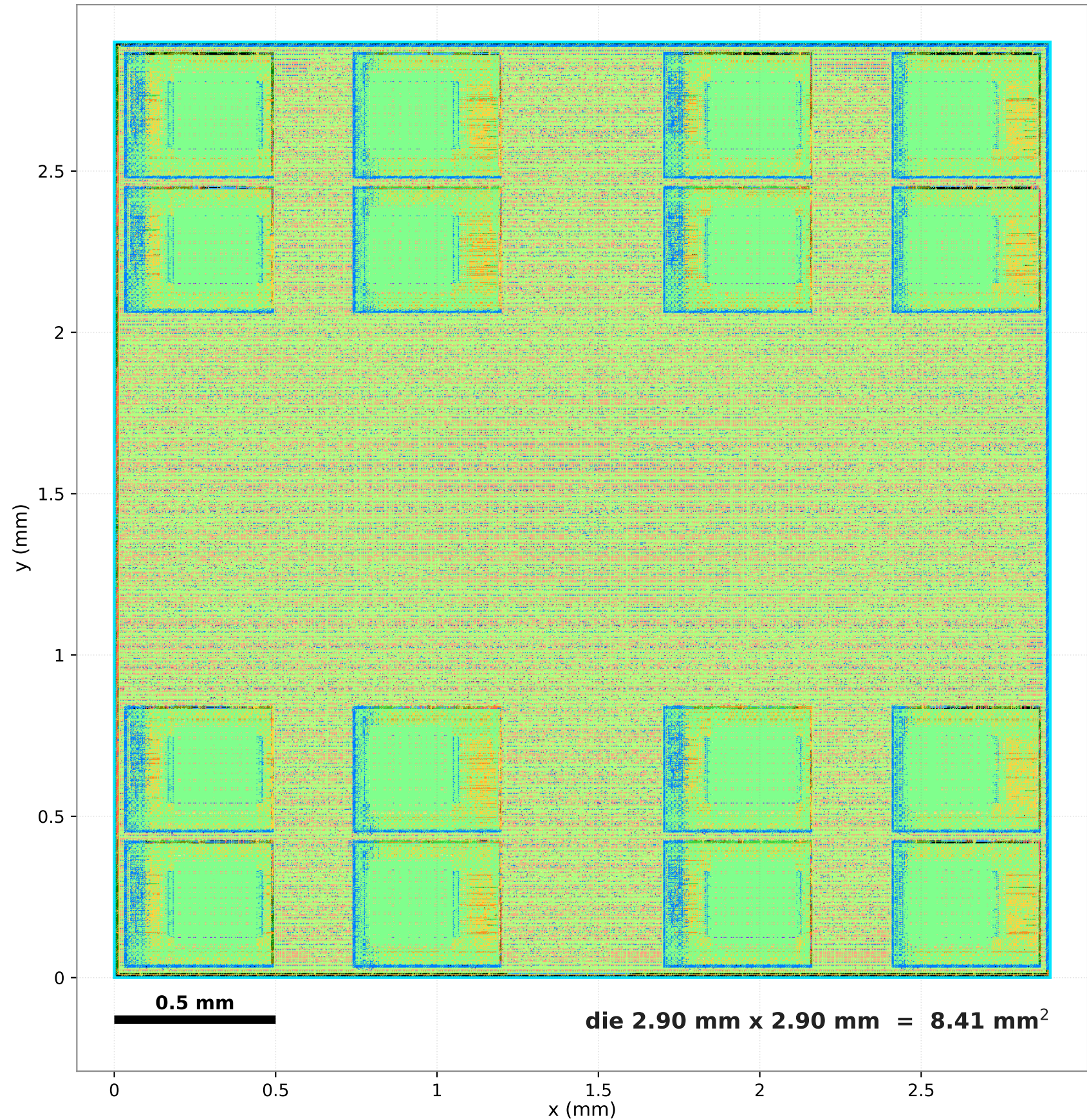


iter26b: 16 KB 4-way non-blocking cache, SKY130 HD, 16 SRAM macros, 188 MHz signed off
GDSII layout (Cache_16384B_assoc4_sram.gds) viewed in KLayout 0.30.7



SIGNOFF SHEET

design/violation numbers from Cadence Innovus 21.16 reports;
timing from Cadence Tempus 23.14 (SI, Quantus RC) - see SIGNOFF

DESIGN

Die size	2.90 x 2.90 mm (8.41 mm2)
Process	SKY130 HD std cells
Std cells	196,962 logic +160,931 decap/tap/diode/tie
SRAM macros	16 x 32x256 (16 KB cache, 4-way)
Clock target	5.300 ns (189 MHz)

VIOLATIONS

Route DRC markers	0
Antenna violations	0

TOTAL violations	0

SIGNOFF (Tempus, SI, both corners)

Clock 5.300 ns (188.7 MHz), propagated, SI-aware		
Setup	ss_n40C_1v76, rc_slow +10%, macro x1.5	
Hold	ff_n40C_1v95, rc_fast -10%, macro x0.67	
Quantus SPEF; ref-pin I/O, in 0.70 / out 0.30 ns		
	5.3 ns	5.5 ns
reg2reg WNS	+0.015	+0.010 ns
reg2out WNS	+1.118	+1.319 ns
in2reg WNS	+0.944	+1.144 ns
hold WNS	+0.075	+0.075 ns
violating paths	0	0
Route DRC 0 / antenna 0 (Innovus verify)		
KLayout BEOL DRC: 9 items outside macros, all known macro-edge / via-master artefacts		
LVS (magic+netgen): 197,438 = 197,438 devices, +64 macro wmask1 tie-off nets only		
Voltus IR: VDD 22 / VSS 22.6 mV worst (limit 52.8)		
Voltus EM: 0 elements over limit		
GLS SDF @5.3 ns: Test1-10 PASS, 221,500 requests		